

Chapter 6

Designing Mathematics Tasks: The Role of Tools

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6.1 Introduction

A mark of human intelligence is our ability to create and use tools to extend our abilities to achieve tasks that are otherwise only unimaginable and to create and use representations to afford interpretation and communication in the objective and personal worlds.

There are different perspectives to interpret tools and representations. Tools can be regarded as mediators between the phenomenological world and the conceptual world, or, according to Radford, “artefacts do much more than mediate: they are a constitutive part of thinking and sensing” (2013, p. 8). Hence, our interaction with tools, artifacts, and culture material should be considered as more than auxiliary elements. Tools influence cognition, and for the purpose of this chapter, they impact on mathematical knowledge. Development of mathematical ideas and concepts has been closely associated with development of technology that, according to Abramovich (2001), can be interpreted as cultural tools in contemporary educational practices. Mathematics teaching that involves tools, among them technology and representations, enables teachers to guide students to (re)invent and visit mathematics.

Children should repeat the learning process of mankind, not as it factually took place but rather as it would have done if people in the past had known a bit more of what we know now. (Freudenthal, 1991, p. 48)

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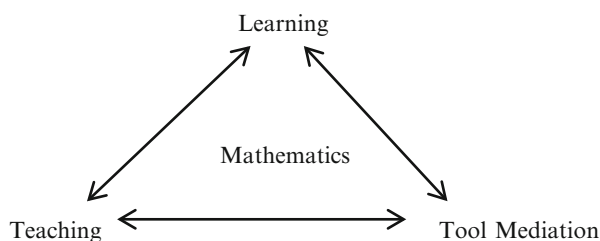
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This chapter concerns designing teaching and learning tool-based tasks in school mathematics. Tools are broadly interpreted as physical or virtual artifacts that have potential to enhance mathematical understanding. A *tool-based task* is seen as a teacher/researcher design aiming to be a thing to do or act on in order for students to activate an interactive tool-based environment where teacher, students, and resources mutually enrich each other in producing mathematical experiences. In this connection, this type of task design rests heavily on a complex relationship between tool mediation, teaching and learning, and mathematical knowledge (Fig. 6.1).

Different epistemological approaches to mathematical learning will have different implications on designing tool-based tasks. For instance, using Sfard's (2008) two metaphors for learning, a *participationist* orientation would favor design with potential for students to participate in the construction of mathematical knowledge/experiences, whereby a more *acquisitionist* orientation would favor design that explores and discovers established mathematical knowledge. Tools can act as a mediator for mathematical discourse; hence, it is important to have views on mathematical discourse in tool-based task design. Two such views are (1) by observing student's discourse, a teacher may find what is missing from the mathematical discourse legitimated by the mathematics community and (2) mathematics is a discourse, so learning mathematics is participating in discourse (Sfard, 2008). These two orientations on the nature of mathematical discourse are relevant considerations for task design because both consider mathematics to be discursive: in (1), students acquire knowledge by learning a specific mathematics language/content; in (2), because mathematics is a discourse, learning mathematics is participating in this discourse, and learning is conceived as changing participation (Rogoff, 1998).

The choice of tool for pedagogical purpose can be considered as a function of how mathematical knowledge is perceived epistemologically by the teacher. The same tool can be used in two task designs that are at opposite epistemological poles. For example, using compasses and ruler may be seen in different ways: for students to construct their own geometrical models in order to explain a certain mathematical phenomenon that they experienced or for students to follow a teacher-given construction procedure to check the validity of a given theorem. An exemplar illustration of the former can be found in the work of DiSessa, Hammer, and Sherin

Fig. 6.1 A relational *triangle* highlighting the interrelationships among different components in a tool-based pedagogical environment



(1991) where a cooperative activity of a sixth-grade class focused on inventing adequate static representations of motion (graphing) using a computational medium, such as Boxer (the Berkeley Boxer Project), was described. The findings indicated students possessed strong meta-representational competence. Tools have potential to empower students to hurdle over didactical and epistemological obstacles (Brousseau, 1998). A tool-based task design could harvest this power to shorten any *distance* between students' prior mathematical experiences and the intended mathematical knowledge to be learned. Rabardel's (1995) theory of *instrumental genesis* was an explication of how usage of a tool can be turned into a cognitive instrumentation process for knowledge acquisition or construction. This approach sees artifacts as psychological tools in a utilization process. It focuses on how a learner develops a scheme of usage for a particular tool while she/he is using it to solve a problem. Pedagogically, this scheme can be attached to the tool to make it into an instrument for teaching and learning. A Vygotskian approach would see artifacts as psychological tools in the context of social and cultural interaction developed through the zone of proximal development and internalization processes (Vygotsky, 1978, 1981). A concrete tool can be transformed into a mind tool through a carefully designed pedagogical environment. In particular, internalization is a socio-semiotic process.

With these constructs as backdrop, Bartolini Bussi and Mariotti (2008) proposed the notion of a *tool of semiotic mediation* for the mathematics classroom, taking on multiple pedagogical functions. On one hand, personal meanings are related to the use of the tool while students accomplish a task; on the other hand, mathematical meanings are related to the tool and its use. This dual relationship constitutes the semiotic potential of a tool. An artifact is regarded as a tool of semiotic mediation if the teacher, intentionally using it, mediates mathematical content through designed didactical intervention. Thus, teachers and students play critical roles in the process of tool mediation. In fact, the teacher, the students, and tools are parts of the mediation process (c.f. Fig. 6.1). Thus, the semiotic potential of tools can be a principle consideration in tool-based task design.

TELMA (see Bottino & Kynigos, 2009) and ReMath projects (see Artigue & Mariotti, 2014) are tool-based task design examples, and we shall describe their work in more detail later. The Technology Enhanced Learning in Mathematics (TELMA) project focused on representation systems and learning contexts. The ReMath Project was a follow-up which produced an academic description of cross-case studies of classroom activity emerging from the implementation of pedagogical plans using digital media.

Other important tool-based task design considerations are how a piece of mathematics can be embodied in a tool and how feedback in tool usage can be used pedagogically. These considerations relate to actual tool manipulation skills and practices; these evolve into amplified learner abilities to discern critical features in a mathematical situation and finally foster the development of mathematical discourses that may or may not be tool dependent.

Confrey et al. proposed that

we need engaging environments, in which the mathematics is actually needed for students to achieve goals that they find compelling, and made visible to students and expressed in a language with which they can connect. (2010, p. 20)

In order to accomplish these concerns, they presented four major thematic approaches for software design: *modeling*, *project-based instruction*, *learning progression*, and *microworlds* (as proposed by Seymour Papert). They present two design examples, Graphs ‘n Glyphs animation and Lunar Land Game, that followed those recommendations. Knowing the rationale behind the design of a tool is an important consideration for tool-based task design as these rationales may form pedagogical bases to harvest the mathematical potential of the tool. A powerful illustration of these connections is presented by Yerushalmy (2015, Chap. 7, this volume).

This introduction sets a stage to delve into a comprehensive exploration of the questions and matters raised. This chapter will present theories, issues, and cases concerning the use of tools in mathematics task design presented and discussed in the ICMI Study 22 Conference Tools and Representation Theme Group (Margolinas, 2013) and other state-of-the-art research. In doing so, we indicate and exemplify heuristics or principles (theoretical or pragmatic) for tool-based task design that are conducive to teaching and learning of mathematics.

In this chapter, we discuss the epistemological, mathematical, representational, and pedagogical considerations of tool-based task design. We then present various theoretical frames and heuristics and give examples of tasks throughout.

6.2 Considerations in Designing Tasks that Make Use of Tools

6.2.1 *Epistemological and Mathematical Considerations*

Epistemological and mathematical considerations play central roles in task design. Different epistemological approaches to mathematical knowledge have different implications on task design. Sfard’s participationist epistemological orientation would favor a tool-based design with the potential for students to participate in the construction of shared mathematical experiences or discourses, whereby the acquisitionist epistemological orientation would use tools to explore and consequently construct personal mathematical knowledge (c.f. Sfard, 2008). The same tool can be used in task designs with different epistemological stances. Thus, tool-based design should couple with how a piece of mathematics content can be learned under a preferred epistemological disposition.

A challenge to tool-based task design is to determine the possible range of epistemological orientation and the type of mathematical knowledge that a tool can

afford and to choose them appropriately for pedagogical situations. Dynamic digital tools, like dynamic geometry software, can be used in task design to cover a large epistemic spectrum from drawing precise robust geometrical figures to exploration of new geometric theorems and development of argumentation discourse. Non-digital tools, such as a set of transparent grid papers with different grid sizes, can be used to design tasks for learning the concept of area, arithmetic operation, etc. depending on how the teacher who designs the tasks understands how the mathematics involved can be manifested and taught using grid systems. Epistemological and mathematical considerations should be closely linked with the tool's potential to represent or manifest mathematical knowledge. Ironically, tools themselves have different degrees of epistemological obstacle through the constraints and limitations of their use, but these that can be used to create cognitive conflict to stimulate the learning process (see Chap. 2 for discussion about cognitive conflict). When thinking of obstacles to learning, there are two possibilities: one is a teaching gap in classroom practices, and the other is a distance between learner's prior mathematical knowledge and the intended mathematical knowledge to be learned. Brousseau (1998) called these, respectively, a *didactical obstacle* and an *epistemological obstacle*. Epistemological obstacles are those knowledge gaps which students need to overcome by construction of new knowledge (Joubert, 2013). Using the notion of epistemological obstacle as a design consideration for a tool-based mathematics classroom should enhance the knowledge construction process because tools can be used as a concrete bridge between students' prior mathematical experiences/knowledge and the intended mathematical knowledge to be learned. For example, the use of a dynamic graphing tool could help students hurdle over the hidden relationships among the a , b , and c parameters in a quadratic equation. A visualization tool can be powerful to overcome many epistemological obstacles in mathematics learning. There is a strong connection between the ways in which students will talk about their mathematical activity and the tools they are using.

6.2.2 Tool-Representational Considerations

Acquiring or constructing mathematical knowledge can be described in terms of creation of representations. The question of the way a chosen tool represents mathematical knowledge is at the heart of tool-based task design. School mathematics is basically symbolic in nature, which is a linguistic type of representation. There are at least two tool-representational considerations that are of design interest for a mathematical topic. The first one is how far away from the expected symbolic representation is in the tool's potential to represent the mathematical concept? Consideration of this distance could become the main concern in the task design approach. This is of particular interest in the use of simple and basic tools that leave ample room for a teacher-designer to exercise his/her pedagogy.

A second consideration is more philosophical and open-ended in nature. Is the tool "capable enough" of representing the targeted mathematical knowledge parallel

to the corresponding symbolic representation, and if so, in what sense? That is, can a tool embed or represent a particular mathematical concept? In this case, the task design is not just to shorten an epistemological distance, but to propose a possible alternative tool-based representation that is compatible with conventional symbolic representation. This second consideration becomes more relevant when emerging powerful ICT pedagogical environments are extending human cognitive abilities and becoming more in alignment with human intellect. For example, blind students can learn geometry through a digital sensory tool via haptic representation of shape. In the work of Healy, Fernandes, and Bolite-Frant (2013), a special tool was designed and implemented with appropriate tasks for blind students to learn about functions (Fig. 6.2). The tool was a digitally controlled board made up of a rectangular matrix of pins, each of which represented a point on the plane. When a particular point is requested or a graph of a given function plotted, the relevant pins are raised up (sequentially as the value of the independent variable increases in the case of the graph of a function), allowing the student to feel, to sense, the image as it is produced (Fig. 6.2). For blind students, this tool-dependent “sensory feeling of the pins” may become their understanding of the concept of function. The tool influences the way the concept is initially understood.

6.2.3 Pedagogical Considerations

Tool-based task design must be supported by a suitable pedagogical environment. Different types of tools afford different mathematical task activities and discourses and have constraints that can be capitalized in constructive ways. Designers and teachers need to take into account possible epistemic interactions among different

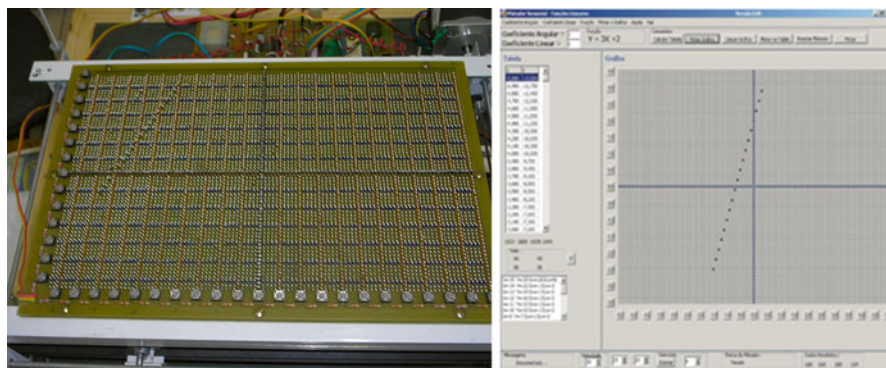


Fig. 6.2 Digitally controlled board made up of a *rectangular matrix of pins*, each of which represents a point on the plane (Healy et al., 2013, p. 67)

types of tools and resources in a pedagogical situation. In Bruner's theory of instruction on the cognitive development of children (Bruner, 1966), three modes of representation were proposed: *enactive representation* (action based), *iconic representation* (image based), and *symbolic representation* (language based). These three instructional modes and the possible cognitive sequences connecting them can serve as a model to guide the design of tool-based tasks that involve different types of tools and classroom resources. With action-based tools, students gain mathematics experiences like number sense or spatial sense through physical sensory means (e.g., Dienes' blocks). Image-based tools lead students to interpret mathematics through visual (dynamic or static) reasoning (e.g., graphics calculator), and linguistic-based tools are conducive to development of mathematical discourse (e.g., commands in CAS tools). Different combinations of Bruner's instructional modes can serve as pedagogical guidelines for tool-based task design. Recent development in ICT environments, like GeoGebra, combined these three modes into one single dynamic multi-representational tool. The enactive and iconic modes connect through the relationship between the mouse, or other driving apparatus, and what appears on the screen. The language used might refer to moving an on-screen object, rather than moving the mouse. Recent developments with touchpad technology connect actions, imagery, and language even more closely to each other and to the intended mathematical structure (Sinclair, 2013; Sinclair & Pimm, 2014). This opens up a rich pedagogical space for task design to explore.

Familiarity with a tool and how to use it effectively to teach and learn are important pedagogical considerations for tool-based task design. Rabardel's (1995) tool-ergonomic theory of *instrumental genesis* focuses on how a learner turns the tool into an instrument by developing a *utilization scheme* that can be mentally attached to the tool and frames how it is used. Furthermore, a tool can progressively be instrumentalized for specific uses by loading it with different potentials through a range of tasks (c.f. Artigue, 2002). Through well-designed utilization tasks, a concrete tool can be internalized to become a psychological tool. A tool can be placed at the center of a pedagogical situation as a means to create a *didactical situation* (Brousseau, 1998) or *didactical intervention* (Bartolini Bussi & Mariotti, 2008); it need not be an adjunct to the didactical event, but can motivate and shape the event itself. Task design should consider the mediation potentials of the tool used in terms of bridging any gaps between the phenomenal world and the conceptual world, the teacher's and students' perceptions of mathematical knowledge, and other pedagogic gaps. Morgan, Mariotti, and Maffei (2009) discuss epistemological and social distances between representations in computational environments:

Distance between representations in different media may be epistemological, affecting the nature of the mathematical concepts available to students, or may be social, affecting pedagogic relationships in the classroom and the ease with which the technology may be adopted in particular classroom (p. 241)

This distance concept can be extended to any tool-based environment and should be an important pedagogical consideration for task design.

6.2.4 Discursive Considerations

Sfard's (2008) commognitive view sees mathematics discourse as a type of communication with special features: keywords, visual mediators, routines, and endorsed narratives. Tool usage can be a means to facilitate the development of these features through task design. Leung (2011) proposed a nested sequence of three epistemic modes in tool-based task design consideration to foster the expansion of a learner's cognitive space: skills and practices, critical discernment, and situation discourse. Practicing to use a tool to accomplish a task involves formation of appropriate tool-based vocabularies in the development of utilization routines. The tool then becomes a mediator for discernment of critical features of the mathematical ideas being sought in the task. Afterward, a tool-based narrative can be constructed to explain the mathematics. This progressive task design epistemic sequence can be seen as a tool-based mathematical discourse development in the commognitive sense. This echoes Radford's idea of a tool as a "constitutive part of thinking and sensing". An overarching question to address is *how to design tool-based tasks that can bring about (situated) discourses for mathematical knowledge mediated by tools in the mathematics classroom, and furthermore, how do these discourses relate to mathematics knowledge?*

6.3 Theoretical Frames for Designing Tool-Based Tasks

6.3.1 Didactical Theories

How can using a tool open up a space of learning where a learner makes epistemic strategies or choices from feedback while using the tool, and what is the role of interaction in this usage? In the Theory of Didactical Situations (Brousseau, 1998), knowledge "is a property of a system constituted by a subject and a 'milieu' in interaction" (Mackrell, Maschietto, & Soury-Lavergne, 2013, p. 80). Balacheff and Sutherland (1994) described *milieu* as "the specific part of the environment of the learner which is accessible and relevant to his or her actions" (p. 140). Interaction can be understood as "a 'dialogue' between the student and the feedback from the milieu" (Joubert, 2013, p. 72). *Milieu* can be considered as a space of learning constituted by didactical variables. A *didactical variable* is a pedagogical parameter that opens a dimension in which students engage in epistemic interaction and which takes on different values depending on student responses to feedback. Thus, the usage of tools in a pedagogical situation can be designed within a milieu in such a way as to create different didactical parameters that are conducive to learning mathematics. In particular, tool-based tasks can be designed to create didactical situations in which students need to construct mathematical knowledge themselves to resolve problems posted by the task.

In Chevallard's Anthropological Theory of the Didactic (described more fully in Chap. 2), mathematics is seen as a product of human activities into which he introduced the notion of an *ostensive* (Chevallard, 1994). An ostensive is "any object

which can be handled concretely by the body, the voice, the vision” (Wozniak, 2013, p. 142). Non-ostensives are “objects” which can only be evoked through the handling of associated ostensives. This implies that mathematical conceptualizations are consequences of acting on and doing using artifacts or tools. Thus, didactically, techniques of using a tool create a dialectic between skills and understanding. Artigue (2002) called these two faces of tool techniques *pragmatic* values and *epistemic* values:

A technique is a manner of solving a task and, as soon as one goes beyond the body of routine tasks for a given institution, each technique is a complex assembly of reasoning and routine work. I would like to stress that techniques are most often perceived and evaluated in terms of pragmatic value, that is to say, by focusing on their productive potential (efficiency, cost, field of validity). But they have also an epistemic value, as they contribute to the understanding of the objects they involve, and thus techniques are a source of questions about mathematical knowledge. (p. 248)

These two values are framed in terms of praxeology: praxis for pragmatic value and logos for epistemic value. “The praxis component contains techniques to achieve a kind of tasks. The logos component includes the theoretical discourses... that describe, explain, justify or develop the techniques used...” (Wozniak, 2013, p. 143). Therefore, a tool can be designed as a didactic variable in a task that forms part of a learner’s space of learning; this variable takes evolving pragmatics and epistemic values which have the potential to transform a learner’s mathematical experiences from experimental to theoretical.

Because tools are potent with pragmatic and epistemic values, designing a tool-based task needs to pay attention to the interplay between techniques and discernment of mathematical concepts and to those features of the tool that have epistemic value (Thomas & Lin, 2013). Kieran et al. (2006) reported a study on the dialectical relation between theoretical thinking and technique, as they co-emerge in a combined computer algebra system (CAS) and paper-and-pencil environment. In their study, the anthropological view was seen as a Task-Technique-Theory triad. A three-part task activity sequence (seeing a pattern, refining a generalization, and proving) was designed based on a CAS and paper-and-pencil environment for students to explore the factoring of $x^n - 1$. The conclusion was that CAS techniques, together with paper-and-pencil techniques, were found to be significant in the deepening of students’ theoretical thinking. Making sense of the CAS outputs and coordinating these with both theoretical notions and paper-and-pencil techniques were fundamental processes in the students’ theoretical and paper-and-pencil progress. This finding supports the existence of a complex pedagogical relationship between the pragmatic and epistemic values of tool use.

6.3.2 Instrumentation Approach to Tool Use

Verillon and Rabardel’s (1995) seminal paper on instrumental approaches to relate cognition and artifacts opened up a vast research arena in the context of using technology in the mathematics classroom (see, e.g., Gueudet & Trouche, 2009;

Trouche, 2004, 2005). An instrument is a psychological tool resulting from instrumental genesis through a learner using the tool to accomplish tool-based tasks. Instrumental genesis is a combination of two processes. Instrumentation is the techno-centric process by which the tool “prints its mark on the subject, i.e., allows him/her to develop an activity within some boundaries” (Trouche, 2004, p. 290). That is, the user is mastered by the tool rather than the other way round. Instrumentalization is the process through which a user develops his/her own ideas of what the tool is designed for and how it should be used. The psychological component of the instrument is the utilization scheme developed by the user (see examples involving TI-92™ in Trouche, 2004; Artigue, 2002). It is usually not easy to distinguish between instrumentation and instrumentalization due to the complexity of the user-tool interactions. To further the idea of instrumental genesis, instrumental orchestration and documentational genesis have been proposed to describe how a tool or a system of tools can be integrated into classroom tasks or purposeful mathematical activities (see, e.g., Gueudet & Trouche, 2009; Trouche, 2004, 2005). Gueudet and Trouche (2009) extended the idea of tool (artifact) to “resources” which are tools (artifacts) with the potential to promote semiotic mediation in the process of learning. Geiger and Redmond (2013) explained:

Resources include entities such as computer applications, student worksheets or discussions with a colleague. A resource is appropriated and reshaped by a teacher, in a way that reflects their professional experience in relation to the use of resources, to form a schema of utilisation—a process parallel to the creation of a schema of instrumented action within instrumental genesis. The combination of the resource and the schema of utilisation is called a document. (p. 121)

Instrumental theories concern the development of a utilization scheme, synergistic interactions among tools, and systematic collation of tool usages in a multiple-tool environment.

6.3.3 *Cultural Semiotic Frame*

Based on a Vygotskian perspective on signs and psychological tools (Vygotsky, 1978), the question of how far tool usage in a pedagogical situation carries mathematical meaning is studied using a tool of semiotic mediation framework proposed by Bartolini Bussi and Mariotti (2008). The framework concerns the relationship between accomplishing a task through the use of an artifact (tool) and learning. It considers the crucial role of human mediation under semiotic and educational perspectives in the teaching and learning process through the artifact. Students produce personal signs while using a tool, and the production and transformation of these signs link to their construction of mathematical knowledge. The link between a tool, a task, and mathematical knowledge is called the *semiotic potential* of the tool in the task context (Arzarello, Bartolini-Bussi, Leung, Mariotti, & Stevenson, 2012, p. 107; Bartolini Bussi & Mariotti, 2008, p. 754). Figure 6.3 is a diagrammatic

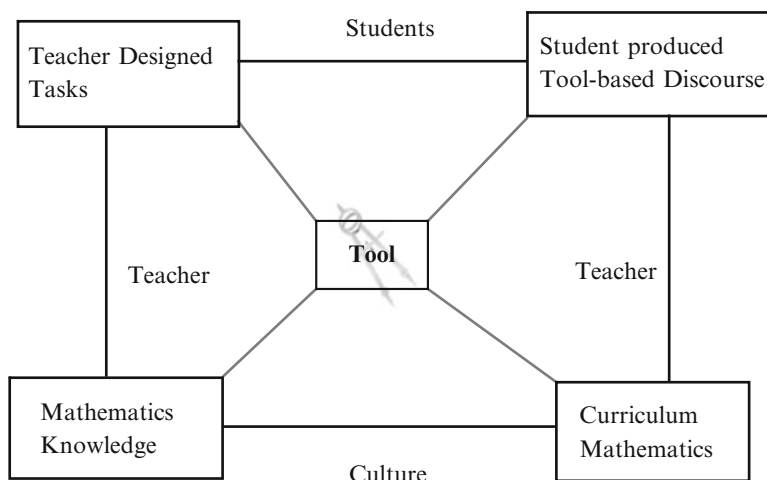


Fig. 6.3 A schematic of the semiotic mediation framework

summary of the semiotic mediation framework, highlighting the interrelationships among the different components of mediation.

Teachers play an important mediator role creating the circumstances in which the mediation process occurs through task design and didactic intervention. Didactic intervention takes the form of didactic cycle “where different categories of activities take place, each of them contributing differently but complementarily to develop the complex process of semiotic mediation” (Mariotti, 2012, p. 29).

Healy, Fernandes, and Bolite-Frant (2013) reported a Brazilian study on how researchers, high school teachers, and deaf/blind students worked together to develop suitable tools to teach and learn matrices. The teacher, who was fluent in Libras (the Brazilian sign language), suggested “that a lack of specific signs for the vocabulary associated with matrices served as a complicating factor in teaching the topic” (*ibid.*, p. 65). Thus, a mediating sign was needed for meaningful feedback to occur. They developed the tool MATRIZMAT, which were boxes that can be joined together by magnets representing rows and columns of matrices. The set for deaf students had numbers put inside the boxes (Fig. 6.4a), while the set for blind students had lids with Braille numbers on them (Fig. 6.4b).

MATRIZMAT became a sensory sign tool (thus “vocabulary” for the deaf and the blind students) that teachers and students used to develop their mathematical language. For the deaf students, new hand gestures were developed to talk about matrices; for the blind students, new spatial positions were explored to experience the meaning of matrices. Thus, feedback generated from a collective (teachers and students) designed tool can be used to develop meaningful classroom mathematical discourse. The semiotic mediation framework gives us a cultural semiotic frame with which to structure tool-based activities that favor production of different categories of signs as mediators for mathematical knowledge.

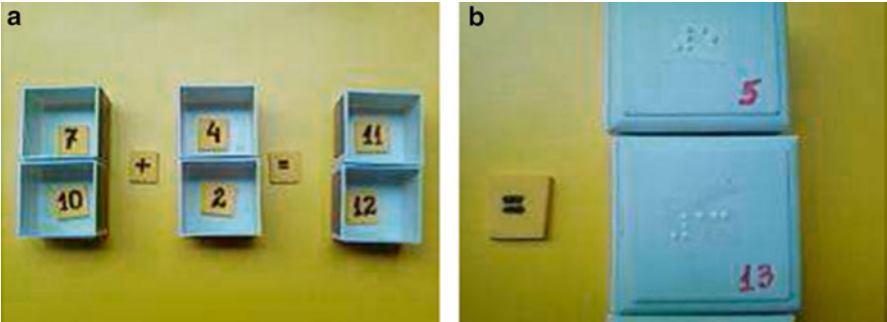


Fig. 6.4 MATRIZMAT in two forms for (a) deaf students with written numerals and (b) blind students with numbers in Braille (Healy, Fernandes & Bolite-Frant, 2013; p. 65)

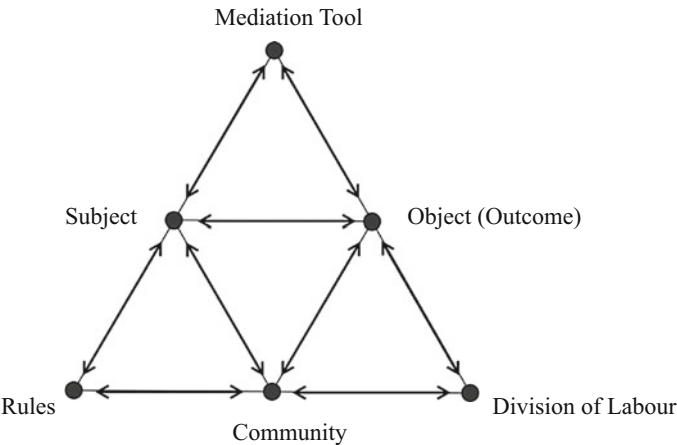


Fig. 6.5 A version of Engeström's activity system (e.g., 1987)

6.3.4 Activity Theory

Another way to look at the relationship between tool use and the pedagogic environment is to look at the interrelationships offered by Engeström (1987) in his structural representation of activity theory. The activity system is seen as an interacting network composed of subject, mediating artifacts or tools, object, division of labor, community, and rules. Figure 6.5 depicts the main components of Engeström's activity system depicted as a mediational triangle.

A tool and its interaction with other components in the activity system play a major role; the arrows in the mediational triangle in Fig. 6.5 serve as guides to design tool-based tasks. For example, the subject is a class of students, the object of activity is to solve an algebraic problem, the tool (mediating artifact) is a CAS, the rules are

how to use the CAS's functionalities, the community is the teacher and students, and the division of labor is organized by pedagogical approaches and arrangements. In this scenario, task design then focuses on how the CAS can create a process of learning comprising of possible connecting activity routes that would lead to possible outcomes.

6.3.5 TELMA and ReMath

Attempts have been made by European mathematics education communities to try to connect different theoretical frames previously mentioned in the context of teaching and learning mathematics using digital tools. From these research projects, important guidelines and concerns have been established relevant to tool-based task design.

TELMA (Technology Enhanced Learning in Mathematics) was a joint collaboration project consisting of six European research teams engaged in cross-experimentation in classrooms using ICT as a tool aiming to develop a common language to analyze the intertwined influence played by different contextual characteristics and theoretical frames (Bottino & Kynigos, 2009). Instead of looking for a unified compromising theory, the TELMA team developed an operational methodology to connect the theoretical frames used by the different teams to study the design and use of ICT tools in the mathematics classroom. In this methodology, the notion of *didactical functionality* (Cerulli, Pedemonte, & Robotti, 2007) provided a common perspective on the use of interactive learning environments in mathematics education, and the Concern Methodology Tool (Artigue, Cerulli, Haspekian, & Maracci, 2009) was used to express key concerns for the didactical functionalities. Three key elements in the definition of didactical functionalities of an ICT tool are (1) tool features/characteristics, (2) educational goals, and (3) modalities of employing the tool in a teaching/learning process. To each element, the Concern Methodology Tool associated a list of concerns: to (1) concerns regarding the ergonomic of the tool, semiotic representation, interaction between student and mathematical knowledge, etc.; to (2) epistemological concerns focusing on specific mathematical contents or specific mathematical practices, cognitive concerns focusing on specific cognitive processes, specific cognitive difficulties, etc.; to (3) concerns regarding the functions given to the tool, instrumental issues, instrumental genesis, etc. (*ibid.*, pp. 221–222). These three functionalities and most of the concerns are relevant tool-based task design considerations that could go beyond the use of the ICT tool. Another key idea developed by the TELMA team was the notion of *distance* (Morgan et al., 2009), in particular, epistemological distance and didactic distance are significant to tool-based task design. *Epistemological distance* refers to distance between different representations (tools) focusing on the difference between the affordances for meaning offered by the representations (tools). *Didactic distance* refers to the nature of and quantity of feedback by the tool asking questions like: *What forms of feedback are provided? How are solutions validated*

and by whom, e.g., by the tool itself, by a teacher, and by peer- or self-validation? (*ibid.*, p. 251). Tool feedback is a key issue in tool-based task design which will be explored further in the later part of this chapter.

The European Research and Development project ReMath (Representing Mathematics with Digital Media) was a continuation of TELMA to “experientially develop and apply strategies for integrating theoretical frameworks and constructs-in-use to enhance knowledge in the uses of digital media for mathematics education” (Kynigos & Lagrange, 2014) based on the work in TELMA. An agreed assumption on a shared idea of (mathematical) representation was the Minimal Theoretical Framework used as a skeleton of implicitly shared theories concerning representation:

1. No direct access to mathematical objects is possible; rather, mathematical meanings are represented through language, formal mathematical notations, and informal idiosyncratic representations.
2. Representations play a fundamental role in the “generation” of mathematical meanings, and this role is assumed to be crucial in the teaching/learning of mathematics.
3. Digital artifacts can provide representations of mathematical objects with a clear potential of generating mathematical meanings (Artigue & Mariotti, 2014, p. 338).

These highlight further the critical role that tools play in learning mathematics and hence the importance of developing tool-based task design.

6.3.6 Other Theories Relevant to Tool-Based Task Design

Mathematics tasks involving the use of tools can be compared with experiments using scientific instruments or construction with tools in the spirit of exploration and invention. The basic tenet of the teaching and learning theory, Realistic Mathematics Education (RME) (Freudenthal, 1973), is described in detail in Chap. 2.

One manifestation of RME is the Freudenthal Institute’s Digital Mathematics Environment (DME), an ICT tool in the form of Java applets designed by teachers, textbook authors, or educators for adapting existing online modules or for designing new ones. It consists of systems to manage and distribute mathematical content and to monitor student progress. Figure 6.6 depicts the DME Authoring Tool interface (Drijvers et al., 2013, p. 55).

Drijvers et al. (2013) illustrated with DME examples how this Authoring Tool with its multiple pedagogical functions supports the three Realistic Mathematics Education (RME) principles that inform task design: guided reinvention, didactical phenomenology, and emergent modeling (*ibid.*, p. 54). These principles point to opening up variant situations and opportunities for students to explore and to investigate and creation of micro-digital worlds (mathematical or not) that are

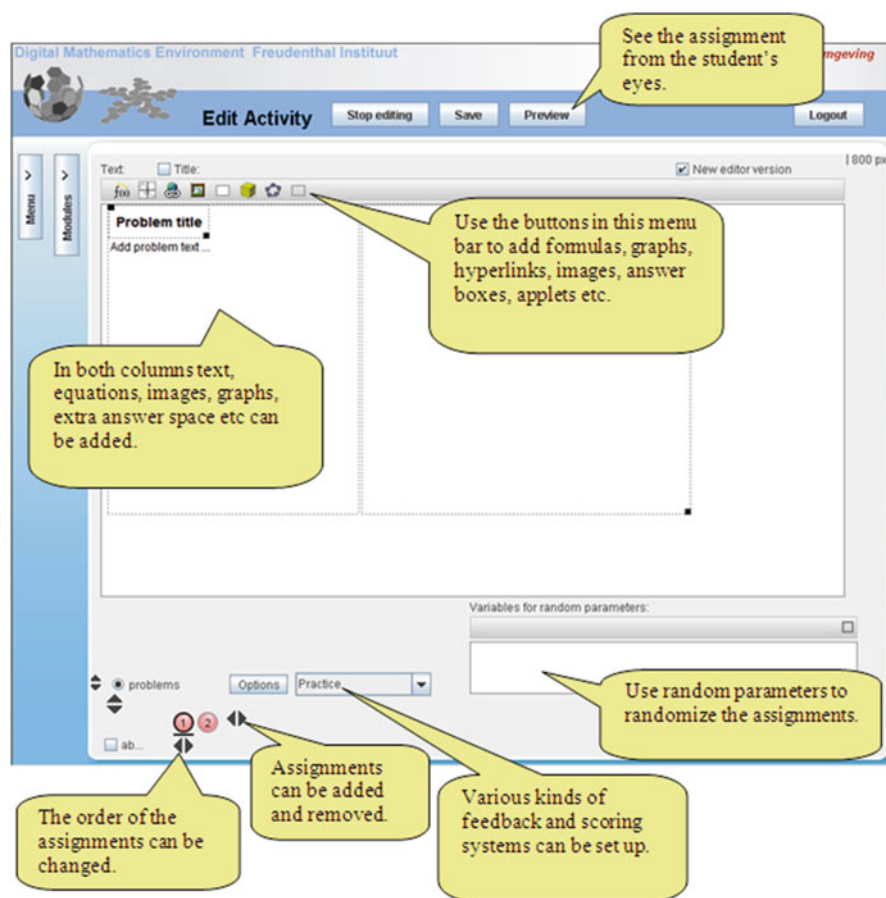


Fig. 6.6 The digital mathematics environment authoring tool (Drijvers, Boon, Doorman, Bokhove, & Tacoma, 2013, p. 55)

real to students. They also point to the evolution of a model developed through different phases of activity by

an increasing repertoire of representations and techniques in the digital environment, or by increasing options to dynamically use these representations, connect them, and switch between them. (Drijvers et al., 2013, p. 5)

When different tools possessing different epistemic values are chosen to put together into a task design, these tools afford different possibilities for learners in ways of “providing them with different types of experiences and activities, as well as different ways to represent ideas and concepts” (Whiteley & Mamolo, 2013, p. 130).

These different ways open up a space for conceptual blending (Fauconnier & Turner, 2002):

Conceptual blending (Fauconnier & Turner, 1998, 2002) is a theory which describes how new inferences can arise when two representations and associated ways of reasoning (or 'input spaces') are brought together in a 'blended concept'. The 'blend' can be thought of as a mapping which combines certain features of the two input spaces and projects them onto a third (newly formed) mental space. (Whiteley & Mamolo, 2013, p. 133)

For example, complex numbers can be seen as a blended concept initiated from two input spaces: real numbers and points in planar geometric space. In this sense, many mathematical concepts are blended concepts. Designing teaching and learning mathematics tasks using multiple tools raises an interesting question asking to what extent a mathematical concept can be regarded as a blended concept inferred from mental spaces generated by the tools. If multiple tool usage has the potential to infer a blended concept in a mathematical activity, then a design consideration is whether or not to design this inference into the task.

Using different tools can create a multi-representational pedagogical environment which provides teacher and student with opportunities to express generality. Clark-Wilson and Timotheus (2013) studied how the multiple representational technological environment (MRT) provided by the TI-Nspire™ can be used to explore mathematical variance and invariance in classroom task design and concluded that

Evidence from the study suggests that the process of designing tasks that utilise the MRT to privilege explorations of variance and invariance is a highly complex process which requires teachers to carefully consider how variance and invariance might manifest itself within any given mathematical topic. The relevance and importance of the initial example space, and how this might be productively expanded to support learners towards the desired generalisation is a crucial aspect of activity design. (p. 51)

An epistemological approach to mathematics is to experience mathematical knowledge through discernment of invariant under variation (see also the description of Variation Theory in Chap. 2). Task design in a tool-based environment opens up pedagogical spaces for exploration of variation and invariant. Leung and Lee (2013) designed a task-based dynamic geometry tool providing a platform to explore variation in students' geometrical perception. Collecting student responses from dynamic geometry drag-based tasks, the platform generated task perceptual landscapes which were used to form and explore students' collective example spaces (see Sect. 6.4.4) and personal example spaces. Variation in the drag-based task design produces different example spaces and hence results in variation in students' geometrical perception. Arzarello, Bairral, Danie, and Yasuyuki (2013) initiated task designs using touch screen multi-figure dragging in the Geometric Constructor (GC) software to study students' perceptions on geometrical concept formation (Fig. 6.7). This new type of dynamic geometry tool will become a new niche to study using variation in task design. Students drag and manipulate dynamic figures with more than one finger on a touch screen.

Special tools can be designed to enhance other abilities in the case of students with a physical disability like blindness (Healy et al., 2013). This tool substitution

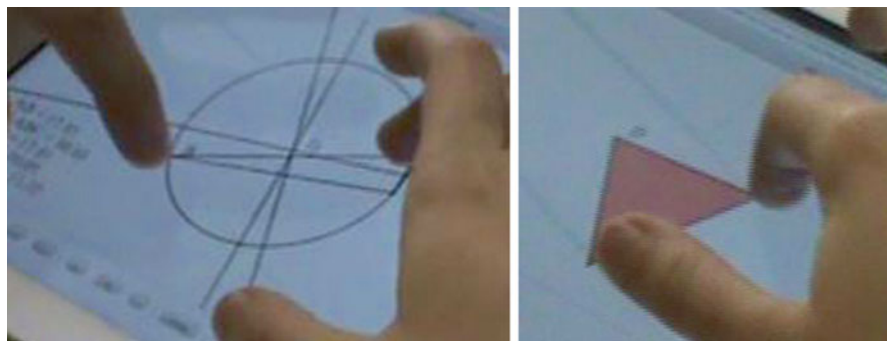


Fig. 6.7 Geometric constructor: a multi-figure dragging touch screen dynamic geometry environment (Arzarello et al., 2013, p. 62)

theory may produce different forms of activity and interaction that could generate different types of mathematical insight and reasoning. “We need to pay attention to—and when necessary create—a multitude of (substitute) semiotic systems to mediate mathematics learning” (Healy et al., 2013, p. 64). Research in tool-based design for inclusion should provide interesting perspectives in mathematics epistemology.

6.4 Further Design Considerations and Heuristics

6.4.1 *Dialectic Between Pragmatic and Epistemic Values*

A gap exists between how to use a tool and how the tool usage leads to mathematics concept development. This gap can be widened if tools are used without meaningful design in the teaching and learning process, especially when skills are involved in using the tools. Without careful consideration on how a skill can be transformed into a cognitive tool to shape a mathematical concept, the use of tools may result in epistemological and didactical obstacles for students, depriving them from reaching the desired or intended mathematical knowledge. The feedback produced by the tool could have negative impact that goes in a direction opposite to the teacher’s intention. Thus, tool-based task design must take into account the pragmatic/empirical-mathematical/systematic gap (Noss, Healy, & Hoyles, 1997) that exists in pedagogical usage of a tool. Joubert (2013) discussed this area of concern in the context of a milieu supported by the computer:

When computers are used, very often the task students are given requires them to construct an artifact on the screen, such as, for example, a graph or a geometric transformation and they begin by working in the pragmatic/empirical field. It seems that frequently the task does not require movement between this and mathematical/systematic fields and mathematical learning is limited. (Joubert, 2013, p. 75)

One perspective to discuss the shortening of the gap is to pay attention to the interaction between the pragmatic and epistemic values of a tool to develop dialectic between mathematical practices and mathematical discourses—*praxeology*:

The teacher must consider the techniques he wants to make alive in the classroom to perform a specific task; he must consider the technological discourses he want the pupils to develop for describing, explaining, justifying and developing the techniques used. (Wozniak, 2013, p. 148)

As the pragmatic value of a tool increases via formation of utilization schemes, its epistemic value may also increase:

The instrumental genesis (Rabardel, 1995) does not facilitate only the use, and thus the instrumentality, of an ostensive. It develops also its semiotic value; an increase of the pragmatic value allows also an increase of the semiotic value. (Wozniak, 2013, p. 148)

Robotti (2013) discussed visuospatial tasks for linear equations using the dynamic algebra artifact AlNuSet to make it possible for students to interpret algebra by a kind of drag-to-fit strategy. For example, Fig. 6.8 depicts snapshots from AlNuSet exploring the meaning of $x + 2 = 2x + 3$. The expressions $x + 2$ and $2x + 3$ are positions on an “algebra line” that depends on the position (value) of x (left figure). Drag x along the algebra line; when it reaches -1 , $x + 2$ and $2x + 3$ automatically line up with each other vertically at 1, which has the same value for the two expressions when $x = -1$ (right figure). This is a dynamic visualization of equality of two linear expressions (Robotti, 2013, p. 104).

In this example, the task was designed to bridge the mathematical conceptual notion “equality” and a discursive notion “drag until they line up”. This discursive notion depends on drag-based visual spatial feedback and forms a dialectal language

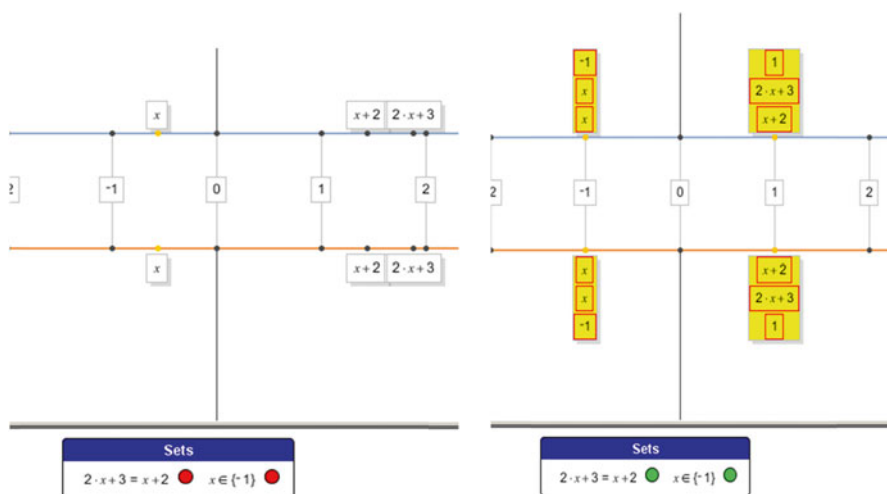


Fig. 6.8 An AlNuSet task exploring the meaning of $x + 2 = 2x + 3$ (Robotti, 2013, p. 104)

in which the pragmatic value of the dragging technique can be transformed into a mathematical concept. The discourse depends on the designing skills of the teacher who is using the software. The AlNuSet also has a Manipulator Component to allow students to solve equations using different symbolic approaches (*ibid*). Combining the algebra line and the manipulator, teachers can design tasks to make a connection between student-produced situated algebra discourse, resulting from the pragmatic value of the tool and the conventional algebraic concepts.

There are further design considerations arising from the dialectic between pragmatic and epistemic values:

- Design cognitive aspects into didactical situations to appropriate the instrumental genesis process.
- Design tool-based activities to explore optimally the epistemic value of techniques.
- Distinguish between, and connect, non-ostensive and ostensive aspects of thinking and doing.

6.4.2 *Mediation and Feedback*

There is another kind of gap that exists in the mathematics classroom, especially in tasks that promote student discourse. Student-produced discourse often differs from what is expected in the mathematics curriculum. One reason may lie in the disparity between mathematical concepts defined by teachers and mathematics experienced by students in the classroom (see Chap. 4). Students usually don't use teacher-defined mathematical concepts in learning activities, but rather use their own taken-for-granted mathematical ideas and discourses in these activities. For example, a teacher defines the notion "perpendicularity" in the classroom while students use a corner of a book to talk about it. To shorten this "discourse gap", a well-designed tool with associated tasks may help, such as a 2-D representation of a 3-D figure in which the corner of a book does not indicate perpendicularity. Tools can become a mediator between different mathematical discourses.

Different types of feedback from tool use can play a major mediation role. In the dynamic geometry environment Cabri Elem, teachers can design their own milieu in which dynamic objects are regarded as didactical variables. Figure 6.9 depicts a snapshot of the opening page in a teacher-designed Cabri Elem activity book (Mackrell et al., 2013). The dots are counters that can be dragged into the different regions of the target scoreboard. When a counter is dragged into the outermost region, the Score will show 1, into the next region will show 10, and into the central region will show 100. Thus, the Score shows the numbers formed by the student. Students can form different 3-digit numbers by dragging counters into the scoreboard in different ways to try to score the preset target number 955. This particular predesigned activity page starts the process of instrumental genesis by dragging counters.

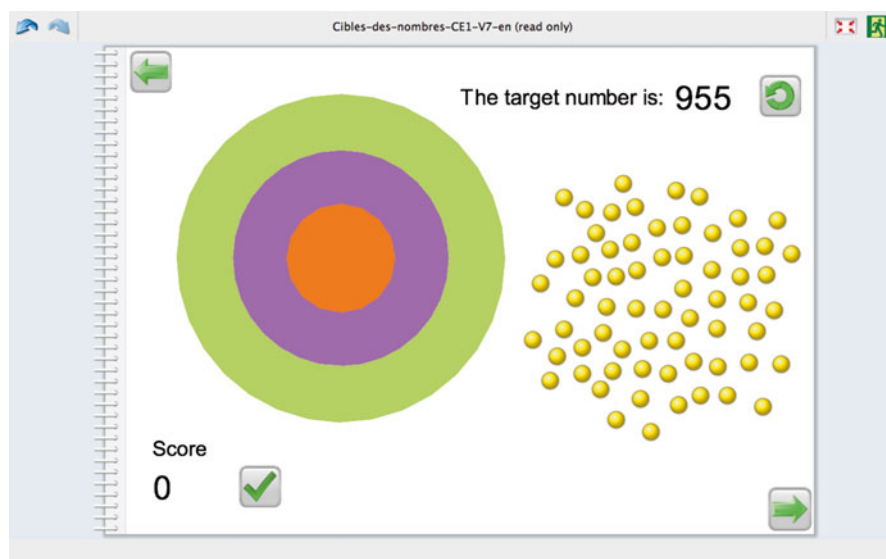


Fig. 6.9 Title page and a task page from the “Target” activity book (Mackrell et al., 2013, p. 83)

The visibility of the score, the movability of a counter, the number of counters, and the target number are all didactical variables that teachers can control. Subsequent activity pages can be designed with different control of the didactical variables in which

an evolution in student strategies may be provoked through changes in the value of the various didactical variables. The ability to choose the way in which pages are linked also enables the provision of optional help in tool use, differentiated tasks, notes for teachers, etc. (Mackrell et al., 2013, p. 83)

This Cabri Elem activity book enables strategic feedback and an interplay between evaluative and direct manipulation feedback (*ibid.*). This evolution can be seen as a kind of *feedback trajectory* that mediates between students’ tool-based discourse and (curriculum) mathematics.

Dragging in a dynamic geometry environment (DGE) affords another important mediation feedback. It is known as soft construction. A soft construction in dynamic geometry is a construction

in which one of the chosen properties is purposely constructed by eye, allowing the locus of permissible figures to be built up in an empirical manner under the control of the student. (Healy, 2000, p. 107)

Constructed by eye means drag a base point of a DGE figure to position(s) that would satisfy a certain condition or property. In the context of task design, Or (2013) used soft construction as a tool to design a GeoGebra task exploring the circumcircle of a triangle using the drag-to-fit strategy. Figure 6.10 depicts snapshots of a soft dragging sequence [(a) to (f)] starting with a predesigned configuration [(a)].

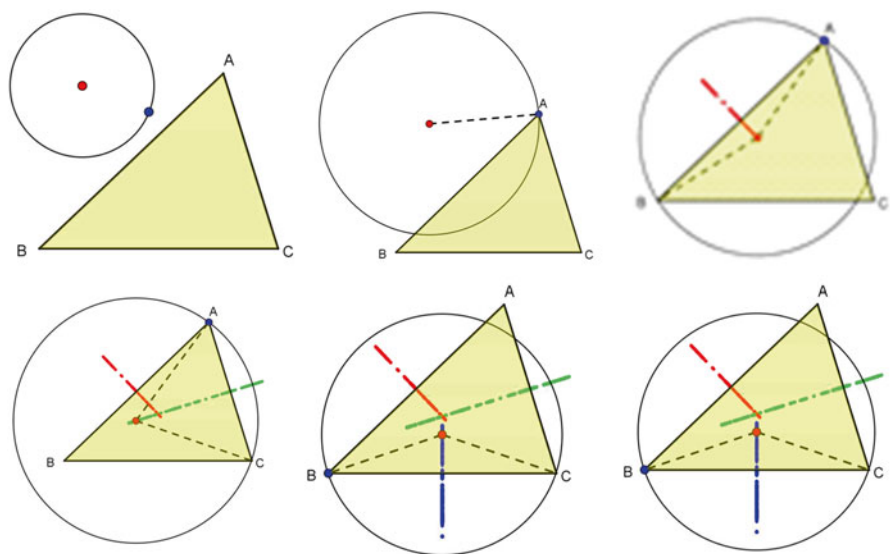
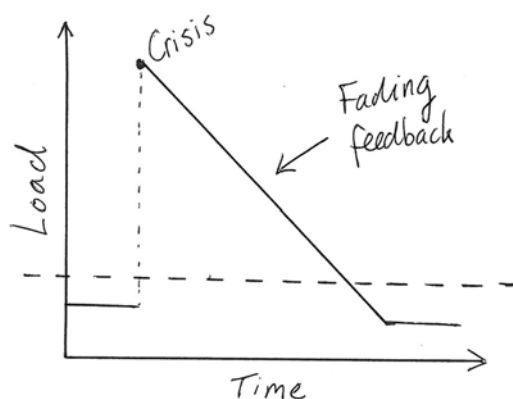


Fig. 6.10 (a–f) Snapshots in a soft dragging exploration sequence of a predesigned DGE task exploring *circumcircle of triangle* (Or, 2013, p. 94)

When the point on the circle is dragged to one of the vertices of triangle ABC , say, A , a dotted radius is shown [(b)]. Soft dragging the center of the circle to positions where the circle passes through another vertex, say, B , another radius is shown and the trace of these positions of the center is marked automatically, resulting in a visual trace [(c)]. At this stage of the task, the teacher can discuss with students the insights gained by what they see on the screen while dragging the center around. Students then continue the discourse by performing the same dragging strategy to the other pairs of vertices [(d) to (f)] and eventually reach the final intended learning outcome [(f)]. The drag-to-fit strategy and the Trace functionality in DGE can be regarded as a kind of recursive feedback that instigates a mediation loop between visualization and reasoning. This soft construction task in DGE is an example of task design making use of this type of feedback.

A specific feedback trajectory can be designed into a task sequence. Bokhove (2013) used a model with elements of *crises*, *feedback*, and *fading* to design a sequence of online near-similar tasks to foster procedural fluency and conceptual understanding in solving algebraic equations. Crisis was intentionally designed into the task sequence at the point where students were beginning to get acquainted with some routine problem-solving strategies. The crisis took the form of modified problems that were similar to what students were familiar with, but may need different problem-solving strategies. Feedback in terms of hints and suggestions was given by the online system to the students in a fading manner as the problem sequence progresses. This model anticipates that students' cognitive load will go through a piecewise decreasing trend in real time with a sharp discontinuous jump at the crisis

Fig. 6.11 A crisis and fading feedback model (Bokhove, 2013, p. 18)



moment (Fig. 6.11). A lowering of cognitive load may imply internalization leading to conceptual development; thus, an online designed feedback trajectory like this one can mediate between a designed discourse and the targeted object of learning, in this case, solving algebraic equations.

The previously discussed cases illustrated different types of tool-dependent feedback that serve to mediate between different aspects in mathematics teaching and learning. Feedback can be regarded as designed or unanticipated didactical intervention. Thus, a tool with designed feedback (actual or potential) can be regarded as a tool of semiotic mediation when used by the teacher to mediate a mathematical content. When teachers design feedback in a tool-based task, they need to anticipate the affordances and constraints of the task as a frame for mathematical activity, types of interaction with the tools, ease of use, uncertainty, available choices, didactic variables, and multi-representation. In terms of mathematics content, feedback acts at the boundary between mathematical and pedagogical fidelity in the sense that the tool-learner interaction creates a bridge between pedagogical design and mathematical meaning. This fidelity boundary can be seen as *discrepancy potential*, which will be discussed in the next section.

6.4.3 *Discrepancy Potential*

In this section, the idea of “distance” that has been manifested throughout this chapter continues with the introduction of the idea *discrepancy potential*. The discrepancy potential of a tool is a pedagogical space generated by (i) feedback due to the nature of the tool or design of the task that possibly deviates from the intended mathematical concept or (ii) uncertainty created due to the nature of the tool or design of the task that requires the tool users to make decisions. For a tool used to teach and learn mathematics, there is an instrumental distance between what the tool can represent and the intended mathematics to be taught. Within that distance,

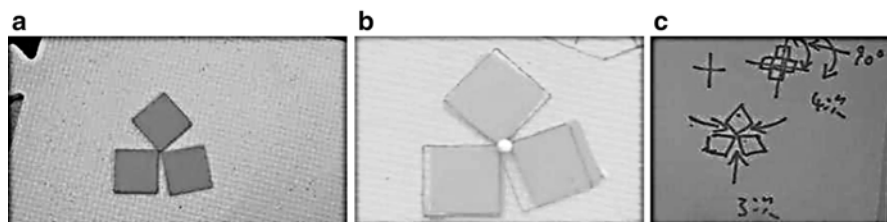


Fig. 6.12 A transparency toolkit used to teach rotational symmetry (Chan & Leung, 2013)

a teacher has the space to introduce disturbances and perturbations, but these might also occur unintentionally.

To illustrate (i), Chan and Leung (2013) reported a study on using a transparency toolkit to teach rotational symmetry in a Primary Five classroom. Students were asked to arrange a rotational symmetric figure using plastic-shaped pieces (Fig. 6.12a). An overhead transparency sheet was placed on top of the figure with a pushpin to pivot it at the center of rotation. Students traced an outline of the figure on the pinned transparency sheet and rotated the transparency sheet to see if the figure satisfied being rotationally symmetrical by paying attention to the overlapping (see Fig. 6.12b).

This was a crude tool and not very accurate. However, because of its crudeness, a pedagogical space was opened for teacher-student discourse that led to development of the mathematical concept. For the episode depicted in Fig. 6.12, the teacher asked students why the overlapping was not exact under the rotation and how to modify the figure to correct this discrepancy. This discussion led to students discovering that for rotational symmetry, the gap angles in the figure must be the same; the more number of times of overlapping in one rotation, the smaller the gap angles become. Afterward, the teacher led the class to explore the angle of rotation for rotation symmetry (Fig. 6.12c). This episode was an unexpected disturbance due to a student-produced figure using the transparency toolkit. The teacher made use of this opportunity to develop a mathematical concept that was not intended in the lesson plan. Thus, a task “which provides an opportunity to make use of the discrepancy embedded in the tool may initiate meaningful mathematics discussion which could lead to deeper conceptual understanding” (Chan & Leung, 2013, p. 41). The notion of discrepancy potential does not have an inherent good/bad value.

To illustrate (ii), uncertainty can be intentionally designed into the task to create cognitive conflict that could bring about discernment:

the potential to *evoke uncertainty and doubt*, which are widely recognized as powerful diagnostic tools as well as vehicles for creating situations in which the need to prove arises intrinsically ... the uncertainty evoked by the tasks trigger students’ discussion and attempts to convince each other through argumentation, which would allow for various aspects of their understanding to be revealed. (Buchbinder & Zaslavsky, 2013, p. 28)

The potential for using discrepancy to foster meaningful mathematical activities can be considered to be a design heuristic. DGE soft construction was discussed in

the last section with respect to mediation and feedback. The epistemic potential of soft construction lies in asking learners during a dragging exploration: *What needs to be in order to get what you want?* It creates a discrepancy potential within which learners can develop and discern mathematical ideas. In particular, relaxing conditions is a heuristic in DGE task design that can probe into different ways to generate conjectures and construct proofs. Arzarello, Olivero, Paola, and Robutti (2002) identified seven dragging modalities (wandering, guided, bound, dummy locus, line, linked, drag test) while trying to analyze conjecture-making episodes by students working on a geometrical problem. The following are two different DGE soft construction task design approaches exploring the cyclic quadrilateral making use of dragging modalities.

Design One

1. Construct a general quadrilateral $ABCD$.
2. Measure two opposite interior angles, say, $\angle ABC$ and $\angle CDA$.
3. Calculate $\angle ABC + \angle CDA$.
4. Turn the Trace function on for point C .
5. Drag point C continuously to keep $\angle ABC + \angle CDA$ as close to 180° as possible.
6. Observe the shape of the path that point C traces out.
7. Make a conjecture on the shape of the path.
8. Explain why the conjecture is true.

Design Two

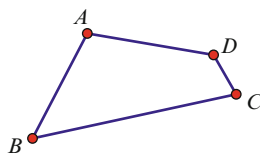
1. Construct a general quadrilateral $ABCD$.
2. Construct a circle that passes through A , B , and C .
3. Construct a circle that passes through D , B , and C .
4. Measure two opposite interior angles, say, $\angle ABC$ and $\angle CDA$.
5. Calculate $\angle ABC + \angle CDA$.
6. Drag the quadrilateral to keep $\angle ABC + \angle CDA$ as close to 180° as possible, and observe what happens to the two circles *or* drag the quadrilateral to make the two circles overlap and observe what happens to $\angle ABC + \angle CDA$.
7. Make a conjecture and explain why the conjecture is true.

In Design One, tracing the point to satisfy the required condition is not obvious. In fact, the trace can be wriggling and hard to discern. However, this discrepancy demands that the learner refines his/her dragging strategies and heighten his/her critical discernment, consequently turning the DGE dragging tool into a mental cognitive tool. Thus, the DGE Trace function can be used (intentionally) to create disturbances that the learner needs to resolve (Fig. 6.13).

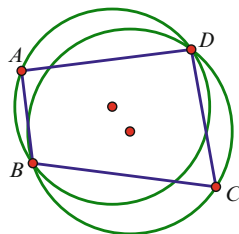
In Design Two, an uncertainty is designed in Step 6 where the learner can choose to take one of the two dimensions of variation (the varying values of $\angle ABC + \angle CDA$ and the “overlappedness” of two circles) as the drag-to-fit target. This intentional designed uncertainty creates a situation where the learner needs to interpret dragging feedback with logical inference sequences and, hence, develops geometrical reasoning (Fig. 6.14).

Fig. 6.13 Design One

$$\begin{aligned}
 m\angle ABC &= 50.19^\circ \\
 m\angle CDA &= 129.57^\circ \\
 m\angle ABC + m\angle CDA &= 179.76^\circ
 \end{aligned}$$

**Fig. 6.14** Design Two

$$\begin{aligned}
 m\angle ABC &= 104.29^\circ \\
 m\angle CDA &= 93.05^\circ \\
 m\angle ABC + m\angle CDA &= 197.34^\circ
 \end{aligned}$$



Possible DGE conjectures resulting from these two tasks are:

Design One

For a quadrilateral to satisfy the condition “a pair of interior opposite angles adds up to 180° ”, the vertices of the quadrilateral should lie on a circular path.

Design Two

Given a quadrilateral $ABCD$, its vertices lie on the same circle *if and only if* a pair of interior opposite angles adds up to 180° .

The idea of discrepancy potential in tool-based task design highlights the importance of paying attention to how the nature or design of a tool can bring about mathematical learning. Different tools can be used to design tasks that foster the teaching and learning of the same mathematical topic in different ways. Their different discrepancy potentials could interact in meaningful ways to bring about rich mathematical experiences for students, as suggested by the overlapping shaded areas in Fig. 6.15. This leads to the issue of multi-representations, which is the discussion in the next section.

6.4.4 Conceptual Blending and Multiplicity

Historically, mathematical ideas have been developed through tools and cultural artifacts. Thus, in retrospect, a mathematical concept should be able to be represented by different tools and artifacts. Furthermore, new and developing technologies play revolutionary roles in representing mathematical concepts and even creating new ones. Multiplicity is a key idea in instrumental orchestration and documentation.

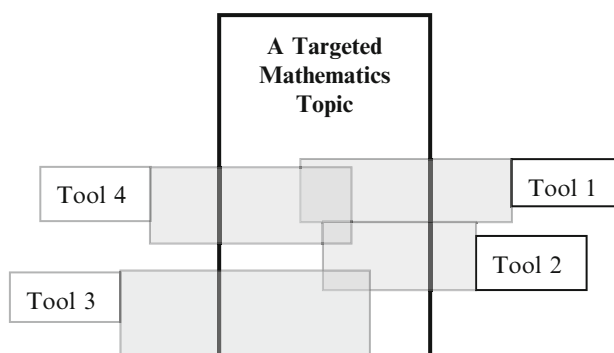


Fig. 6.15 The *shaded regions* are the tool discrepancy potentials for different tools

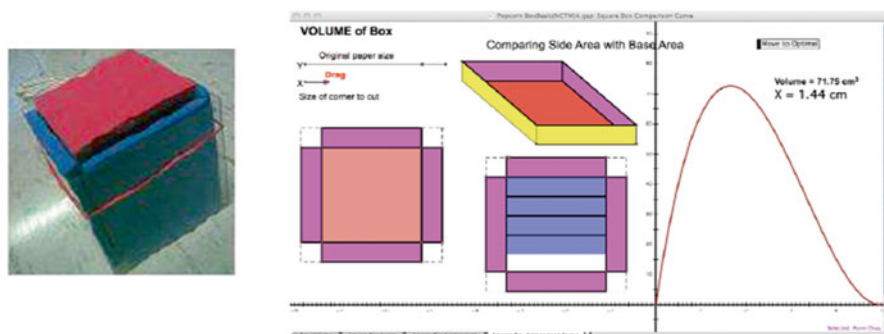


Fig. 6.16 A multi-tool environment consisting of concrete models and interactive 2-D representations of 3-D *boxes* and function graphs in Geometer's Sketchpad (GSP) (Whiteley & Mamolo, 2013, pp. 137–138)

Often, abstract generalizations come about when critical aspects from multiple mathematical representations and discourses fuse and blend together. Therefore, in a multi-tool pedagogical environment, interactions among the discrepancy potentials of different tools, as discussed in the previous section, and the bridging and switching between tools and representations are key task design considerations.

Whiteley and Mamolo (2013) used a framework of conceptual blending (Turner & Gilles Fauconnier, 2002) to design a multi-tool task that deals with a box optimization problem. The multi-tool environment consisted of concrete models and interactive 2-D representations of 3-D boxes and function graphs in Geometer's Sketchpad (GSP) (Fig. 6.16).

It was found that teachers and students had multiple ways of reasoning about the task and created different conceptual blends for these representations. Nevertheless, the selected problem encouraged all participants to anticipate that there was an optimal shape and guess a candidate shape (always incorrectly). Even simple activities could indicate that better reasoning tools were needed to resolve the conflict and

find the optimum. For teachers, the task was about allowing an opportunity to unpack the basic processes of calculus and optimization and to realize a novel spatial sense in optimizing geometrical problems. For students, the task afforded them with accessible tools to solve a problem in different ways, which supported or laid a foundation for a symbolic/algebraic approach (Whiteley & Mamolo, 2013, p. 134). Using a conceptual blending framework to analyze the multi-tool environment and the activity emerging from the task gives information about the discrepancy potentials of the individual tools and also the conceptual development of the students (*ibid.*, p. 136). The concrete model became a thinking tool and the GSP applet invited rapid (or fluent) switching among multiple ways of reasoning which were supported by the multiple representations. Task design should consider the mental processes involved in rapid (fluent) switching back and forth between representations and students' alternative ways of reasoning. With the lens of conceptual blending, individual student difficulties can be tracked to gaps in how particular representations are manipulated to support reasoning. With practice, learners can apply the reasoning elicited by one tool within the context of a second tool, for example, taking the focus on “sign of the change” in the model to the slope of a secant in the graph.

Addressing the issue of students' multiple feedback responses to a tool, Leung and Lee (2013) designed predesigned dragging tasks in a DGE-based platform that was able to record multiple student responses in a collective fashion. The platform was capable of generating a collective image map of student geometrical perceptions for a predesigned dragging task. This map, called a *task perceptual landscape*, is visually interpreted as students' qualitatively different ways of perceiving a geometrical phenomenon under the drag mode, ways which are quantified and categorized in a collective way (see, e.g., Fig. 6.17). The scatterplot on the left consists of students' responses superimposed on the dynamic task template. The task perceptual

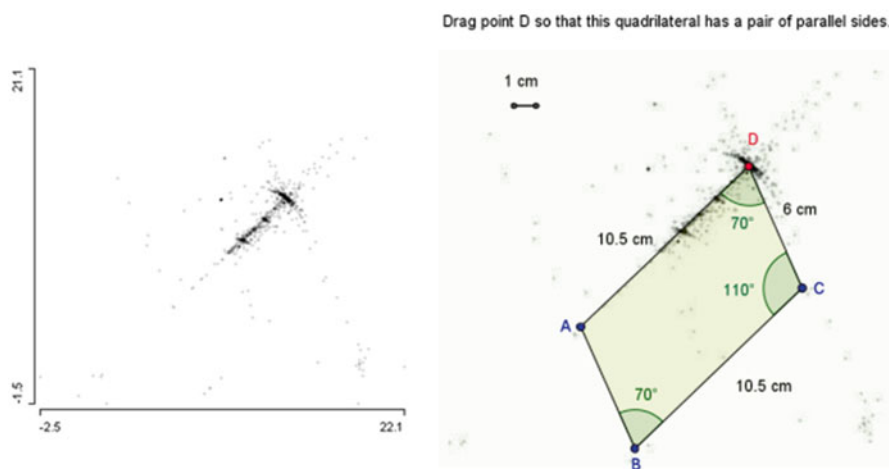


Fig. 6.17 Dynamic task template (Leung & Lee, 2013)

landscape is a *collective example space* of students' perceptions of a task. This collective student example space can be used to design teaching and learning tasks to explore students' perceptions on geometrical understanding. In this case, multiplicity is seen in terms of amalgamation of multiple student responses in a dynamic interactive tool environment. This task-based dynamic geometry platform opened up the idea of an *instrumental documentation environment* that should have potential pedagogical implications which can be explored.

Kaput (1986) postulated that a multiple representational environment supported by technology might enhance high-level engagement with mathematics. In this connection, Clark-Wilson and Timotheus (2013) studied the design rubric of a multi-representational technological environment (MRT) supported by the TI-Nspire. In particular, the study focused on how MRT helped to explore mathematical invariance and variance. A key task design consideration was how to switch and flow different representations (e.g., graphical, tabular, and linguistic) so their interactions would bring about mathematical generalization and how the situated discourses that developed under these representations communicate with each other with respect to the generalization. The findings gave rise to seven questions that are relevant to task design in a multi-tool environment (Clark-Wilson & Timotheus, 2013, p. 51). They take seriously the assertion that "a lesson without the opportunity for learners to express a generality is not, in fact, a mathematics lesson" (Mason, Graham, & Johnston-Wilder, 2005, p. 297). In that sense, these questions apply equally well to all tool-based task design contexts:

1. What is the generalizable property within the mathematics topic under investigation?
2. How might this property manifest itself within the multi-representational technological environment—and which of these manifestations is at an accessible level for the students concerned?
3. What forms of interaction with the MRT will reveal the desired manifestation?
4. What labeling and referencing notations will support the articulation and communication of the generalization that is being sought?
5. What might the "flow" of mathematical representations (with and without technology) look like as a means to illuminate and make sense of the generalization?
6. What forms of interaction between the students and teacher will support the generalization to be more widely communicated?
7. How might the original example space be expanded to incorporate broader-related generalizations?

6.5 Synthesis

For some theories, tools and representations act as extensions to our sensory/mental cognition to experience and perceive mathematics. Tools may start as external artifacts, but they have the potential to turn into internal cognitive tools or cognitive extensions. For embodied theory, in which there is no external/internal dichotomy

Choices of Tool	Visible crises
Open-endedness	Formative assessment
Connectivity	Didactic phenomena
Accessibility	Task sequence design
Freedom to choose	Variation of didactic variables
Immersion in the technology	Perturbation and disturbance
Possibility of student as a designer	Feedbacks: Controlled, uncontrolled,
Constraining or respecting diversity	fading, student-generated, teacher-designed

Fig. 6.18 Issues around pragmatic and epistemic dialectic

and cognition is dependent on the activity, tools are regarded as part of cognition. In either case, whether tools are conducive or disruptive in the process of teaching and learning depends on how they are designed to be used in pedagogical situations. As discussed in the previous sections, choices of tool and design of tool usage open up pedagogical spaces where the pragmatic and epistemic dialectic is at play and where students and teachers are invited to develop mathematical discourses. Figure 6.18 displays some pedagogical issues surrounding the tool-based pragmatic and epistemic dialectic that task design needs to pay attention to.

To focus further on tool-based task design, considerations, and heuristics, there are at least four broad areas of concerns:

1. Use implicit, explicit, and strategic feedback from the tool-based environment to create crises and/or fading that are conducive to student learning.
2. Design pragmatic/instrumental and epistemic/semiotic activities that consider the boundary of mathematical and pedagogical fidelity, tactile and meaningful interactions with the tools, and substitution or enhancement of sensory abilities.
3. Make use of the affordances and constraints created by the tool. For example, unintentional disturbances caused by the tool quality, different design given by relaxing conditions/functionalities, (epistemological) obstacles created by the tool, etc.
4. Fluent switching between representations/tools.

These heuristics should not be seen as separate and distinct design considerations; rather they are various foci of attention in the design process, and they inform and reinforce each other. A few key heuristics discussed in this chapter are further highlighted in the following.

6.5.1 Strategic Feedback and Mediation

Feedback is a key to promote skill and understanding in any type of tool usage. Types of feedback can be regarded as didactical variables that entail possibly different pragmatic and/or epistemic values conducive to mathematical concept development.

Thus, designing appropriate strategic feedback into tool-based tasks should be a major design attention. Feedback always has potential to open up a space for unanticipated actions and/or discourse (especially coming from didactical and epistemological obstacles) with potential pragmatic and/or epistemic values. Task design should be flexible enough to allow development of this contingent pedagogical space. Tool usages can be developed as semiotic signs that can mediate mathematical knowledge. Whether this process is natural, designed, or a combination of both depends on the choice of tool, the mathematical content, and the choice of pedagogical approach. Feedback and mediation are complementary processes leading internalization and/or transformation of knowledge.

6.5.2 *The Instrumental/Pragmatic-Semiotic/Epistemic Continuum*

Evolution of a tool's instrumented action scheme (Trouche, 2004) transforms the instrumental/pragmatic-semiotic/epistemic dialectic. The instrumental genesis and tool of semiotic mediation frameworks (Sect. 6.3) both deal with this transformation from physical tool to psychological tool. Task design needs to sequence tool-based activities to progress through the *epistemological continuum* where mathematical skills/techniques are gradually transformed into abstract concepts and understanding. Tools can be assigned or designed to play different epistemic roles at different points of the continuum. For example, tools can enable: action, formulation, and validation (Brousseau, 1998). Action involves familiarity with the tool and how to implement a procedure to use the tool. Formulation concerns a task situation when tool feedback produces difficult unfamiliar problems for students to tackle that need to be resolved mathematically. It is in this mode that a tool begins to turn into a psychological tool for discernment of critical mathematical ideas. Validation dialectics involve explaining, justifying, theorizing, and proving. This echoes with many aspects in Leung's nested epistemic modes task design framework mentioned in Sect. 6.2.4.

6.5.3 *Boundary Between Mathematical and Pedagogical Fidelity*

Task design can situate a tool at the boundary between mathematics and pedagogy. This is in the sense that a tool can be designed to embody a pedagogical dimension of mathematical knowledge. Thus, the mathematical and pedagogical usage of a tool becomes a task design issue. A pedagogical dimension may be, for example, a didactical situation that is relevant to students' mathematical experiences but may not immediately access to the rigorous mathematics dimension, or vice versa. Task design should be sensitive to this boundary and aim to create a symbiosis between mathematics and pedagogy.

6.5.4 *Discrepancy Potential*

Tools arouse uncertainty and doubt because feedback from tool usage may deviate from the intended mathematical concept to be learned or may create cognitive conflicts for students. This opens up a potential pedagogical space where teacher and students have opportunities to use such discrepancies to develop mathematics discourse that could lead to formation of mathematical concepts. Task design can intentionally make use of a tool's discrepancy potential to create uncertainties and cognitive conflicts which are conducive to student learning.

6.5.5 *Multiplicity*

Different tools extend or amplify different abilities. A multi-tool teaching and learning environment provides learners a milieu where they can interact with different tools and representations. In such a pedagogical situation, task design should pay attention to design connection, switching, and transition between tools and representations to bring about contrasting experiences for learners which may lead to critical discernment of invariant mathematical concepts. Mathematical activities are mostly concerned with discerning invariants in variation or seeing how invariants appear in variation. Thus, designing variation tasks in a multi-tool environment should enrich the mathematics knowledge acquisition/construction process.

As said, the above heuristics are not separate considerations for tool-based task design. They are all closely related to each other and are critical aspects in the whole design process. At present, no overarching or unified theory for tool-based task design exists. The TELMA and ReMath projects experimented with a cross-experimentation and cross-case analysis methodology aiming to create a dialogue platform where different theoretical frameworks were used for task design in an ICT environment; some of these frameworks are discussed in this chapter (Kynigos & Lagrange, 2014). Their findings should serve as a solid reference for tool-based task design. A direction for tool-based task design research is to see how far the findings from these two projects extend to non-ICT tools. The discussion in this chapter is a preamble to this course of research. Technology is developing at an exponential rate, and the nature of reality is becoming less easy to define. Integrating advanced technological tools into the mathematics classroom challenges the nature of mathematical knowledge and, in particular, the question of what represents mathematics. In this direction, tool-based task design principles and heuristics evolve with emerging technology. Therefore, teachers who design tasks should be aware of his/her disposition toward the nature of mathematics as this will influence how the designed tool and its use will be seen to represent mathematics.

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